

Suggested Teaching Guideline for
Programming Concepts - PGCP-ITISS February 2026

Duration: 30 Theory hrs + 40 Lab hrs+ 20 Self-Learning hrs (Total: 90 hrs)

Objective: Develop students with the ability to efficiently automate tasks, manage data, and develop custom solutions using Python and databases.

Prerequisites: Basics of Computer Fundamentals and Database

Evaluation method: Theory exam- 40% weightage
Lab exam- 40% weightage
Internal Assessment - 20% weightage

List of Books / Other training material

Courseware: Learning with Python by Allen Downey/ wiley, 2023

Reference: Python for Unix and Linux System Administration

A. MySQL: (10T + 12L + 8 SL)

Session 1: (2T + 2L + 2SL)

Theory:

- Concepts of databases, tables, and schemas.
- Syntax and examples for creating databases.
- Best practices for organizing and naming databases.
- Syntax and examples for dropping databases.

Lab Assignment:

- Create a MYSQL user.
- Create a database add tables and then add data into those tables.
- Delete/Modify a Table or Database.

Self-Learning:

- Explore the topics
 - ◆ Concept of ER Diagrams
 - ◆ Entities, Attributes, Primary key, Foreign key
 - ◆ Mapping ER diagram -> Mysql tables
 - ◆ Draw an ER Diagram and convert it into CREATE TABLE statements

Session 2 & 3: (4T + 6L + 2SL)

Theory:

- Writing SELECT statements to fetch data.
- Using WHERE clause for filtering data.
- Sorting data using ORDER BY.
- Basic JOIN operations to combine data from multiple tables.

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Lab Assignment:

- Write SELECT statements to retrieve specific data from a database.
- Use WHERE clauses to filter data and extract relevant information.
- Apply ORDER BY clauses to sort retrieved data for analytical purposes.

Self-Learning:

- Explore the topics [Database Security in MYSQL]
 - ◆ Unauthorized access
 - ◆ Data leakage
 - ◆ Concepts of SQL Injection
 - ◆ Users, Roles, Privileges
 - ◆ Principle of least privilege
 - ◆ Create multiple MYSQL users: admin, readonly_user, app_user
 - Assign privileges using: GRANT,REVOKE

Session 4 & 5: (4T + 4L + 4SL)

Theory:

- Using aggregate functions (SUM, COUNT, AVG).
- GROUP BY and HAVING clauses for grouped data.
- Subqueries and nested queries.

Lab Assignment:

- Utilize aggregate functions (SUM, COUNT, AVG) to perform calculations on dataset columns.
- Employ GROUP BY clauses to group data based on specified criteria.
- Use HAVING clauses to filter grouped data based on aggregate conditions.

Self-Learning:

- Explore the topics [Database Security in MYSQL]
 - ◆ Unauthorized access
 - ◆ Data leakage
 - ◆ Concepts of SQL Injection
 - ◆ Users, Roles, Privileges
 - ◆ Principle of least privilege
 - ◆ Create multiple MYSQL users: admin, readonly_user, app_user
 - Assign privileges using: GRANT,REVOKE
- Explore the topics [Backup, Restore & Disaster Recovery]
 - ◆ Importance of backups in production systems
 - ◆ Learn the concepts of Logical vs Physical backup
 - ◆ Restore process & common failure scenarios
 - ◆ Delete data/table intentionally and restore from backup
 - ◆ Case Study: What happens if production database crashes at midnight.
- Explore the topics [Cloud Databases & Performance Basics]
 - ◆ Cloud Database

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- ◆ Differences between On-Prem MYSQL vs Cloud MYSQL
- ◆ Vertical vs Horizontal scaling
- ◆ Indexing
- ◆ Learn the concepts of slow queries
- ◆ Make a comparison between Local MYSQL vs Amazon RDS MYSQL
- ◆ Identify: Backup handling, Scaling options and Security features offered by Cloud database.

B. Python: (20T +28 L + 12 SL)

Session 6: (2T + 2L)

Theory:

- Installing Python
- Your First Python Program
- Declaring Functions
- Overview of Objects
- Indenting Code
- Native Data types

Lab Assignment:

- Try to install Python in windows and Linux (if by default not installed)
- Program on Datatypes.
- Make small program to print Hello Word
- Make a script file in Python and run that script.
- Take an input from user while running the program and print the output.
- Use for/while loop and If-else condition in program.

Session 7 & 8: (4T +4L +2SL)

Theory:

- Introducing Dictionaries
- Defining Dictionaries
- Modifying Dictionaries
- Deleting Items from Dictionaries
- Introducing Lists
- Defining Lists
- Adding Elements to Lists
- Searching Lists
- Deleting List Elements
- Using List Operators

Lab Assignment:

- Create a dictionary, print that dictionary.
- Use for/while loop while running the program.
- Insert/delete an item in dictionary.

Self-Learning:

- Given an unsorted list of integers nums, find the length of the longest consecutive sequence of elements. The algorithm should run in O(n)

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time.

Example: Input: nums = [100, 4, 200, 1, 3, 2]

Output: 4 (The sequence is [1, 2, 3, 4])

- Given a list of integers nums, return a list output such that output[i] is equal to the product of all the elements in nums except nums[i]. You must solve it without using division and in O(n) time.
Input: nums = [1, 2, 3, 4]
Output: [24, 12, 8, 6] (234=24, 134=12, 124=8, 123=6)
- Given a list of intervals, where each interval is represented as a list [start, end], merge all overlapping intervals and return a new list of the non-overlapping intervals that cover all the intervals in the input.
Example: Input: intervals = [[1,3],[2,6],[8,10],[15,18]]
Output: [[1,6],[8,10],[15,18]] (Intervals [1,3] and [2,6] overlap, merging into [1,6])

Session 9: (2T + 2L + 2SL)

Theory:

- Introducing Tuples
- Declaring variables
- Referencing Variables
- Assigning Multiple Values at Once
- Joining Lists and Splitting Strings
- Introduction to Set

Lab Assignment:

- Create a tuple, print that tuple.
- Use for/while loop while running the program.
- Insert/delete an item in tuple.

Self-Learning:

- Write a Python function swap_tuple_elements(tup, i, j) that takes a tuple tup and two indices i and j as input. The function should return a new tuple where the elements at indices i and j are swapped.
input_tuple = (10, 20, 30, 40)
swap_tuple_elements(input_tuple, 1, 3)
Expected output: (10, 40, 30, 20)
- Given a list of tuples, where each tuple contains a name (string) and a score (integer), write a Python function sort_scores(data) that sorts the list of tuples in descending order based on the score.
scores = [('Alice', 88), ('Bob', 92), ('Charlie', 75), ('David', 92)]
Expected output: [('Bob', 92), ('David', 92), ('Alice', 88), ('Charlie', 75)]
- Given 2 tuples, perform cross pairing of corresponding tuples, convert to single tuple if 1st element of both tuple matches.
Input : test_list1 = [(1, 7), (6, 7), (8, 100), (4, 21)],

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```
test_list2 = [(1, 3), (2, 1), (9, 7), (2, 17)]
```

Output : [(7, 3)]

Explanation : 1 occurs as tuple element at pos. 1 in both tuple, its 2nd elements are paired and returned.

Input : test_list1 = [(10, 7), (6, 7), (8, 100), (4, 21)],

```
test_list2 = [(1, 3), (2, 1), (9, 7), (2, 17)]
```

Output : []

Explanation : NO pairing possible.

Session 10: (2T + 2L + 2SL)

Theory:

- Optional and Named Arguments
- type, str, dir, and Other Built-In Functions
- Object References
- Socket with Python

Lab Assignment:

- Append/insert/extend/remove/sort/reverse a list.
- Append/insert/extend/remove/sort/reverse a string.
- Use Built-in functions.

Self-Learning:

- Given a string, find the length of the longest substring that does not contain any repeating characters
Example: `length_of_longest_substring("abcabcbb")` should return 3 (for "abc").
- Given a string s, return the longest palindromic substring in s.
Example: For "babad", the answer is "bab" (or "aba").
- Given a string s consisting of opening and closing parenthesis '(' and ')', find the length of the longest valid parenthesis substring. A valid parenthesis substring is one where every opening bracket '(' has a corresponding closing bracket ')' in the correct order.
Examples: Input: s = "()"
Output: 2
Explanation: The longest valid parentheses substring is "()".
Input: s = "(()())"
Output: 6

Session 11: (2T + 4L + 2SL)

Theory:

- Regular Expressions Using python
- Scripting in Python
- Functions and Functional Programming

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Lab Assignment:

- Use regular expression to take a regular expression and search any pattern.
- Do all of these with the help of regular expression.
 - ◆ Match a single character except newline.
 - ◆ Match a word.
 - ◆ Match a single whitespace character.
 - ◆ Match any Non-whitespace character.

Self-Learning:

- Write a Python program to check that a string contains only a certain set of characters (in this case a-z, A-Z and 0-9)
- Write a Python program to remove leading zeros from an IP address.
Input: 216.08.094.196
Output: 216.8.94.196
- Write a Python function that converts a CamelCase string into a snake_case string using re.sub(). The solution should insert an underscore before every uppercase letter (except the first one) and convert the entire string to lowercase. Example
Input: "thisIsAConversionExample"
Expected Output: "this_is_a_conversion_example"
- Given a sample log file line, use re.split() to divide the string into a list containing the timestamp, log_level, message, and details. The separators are consistent patterns of brackets, spaces, and hyphens. Example Input: "[2025-12-17 10:00:00] - ERROR - User login failed: Invalid credentials"
Expected Output: ['2025-12-17 10:00:00', 'ERROR', 'User login failed', 'Invalid credentials']

Session 12: (2T + 3L)

Theory:

- Object Oriented Linux Environment
- Classes, Objects and OOPS concepts

Lab Assignment:

- Do practice of memory management questions.
- To find out Current directory of a file
- To find out all folders & files inside given folder.
- To find out absolute path and relative path.
- To check if a file or folder exists or not.
- To create/remove a Directory.
- To check the permissions.

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Session 13 & 14: (4T + 6L)**Theory:**

- File Handling
- Directory Access Permissions
- Controls Socket

Lab Assignment:

- To read or write a file using python.
- Print everything written in a file.
- Count, lines of the file
- Count, words of the file
- To read a web page content

Session 15: (2T + 5L+ 4SL)**Theory:**

- Libraries (eg., hashlib, shodan, datetime)
- Servers and Clients Arch
- Web Servers and Client scripting
- Exploit Development techniques
- Debugging basics

Lab Assignment:

- Create a socket between a server and a client then analyse and monitor all activities of client.
- Exploit using socket programming
- Exploit analysis
- Debugging

Self-Learning:

- Explore about sorting and searching Algorithms
- Learn the difference in time complexity (best, average, worst case)
- Learn about Divide and Conquer Approach
- Make a comparison between divide and conquer Vs Dynamic Programming
- Learn about Number theory
- Learn how bit manipulation works
- Solve the following:
 - ◆ Given an array of integers where every element appears twice except for one, find that single number. Your algorithm should have a linear runtime complexity and use only constant extra space. Input: [4,1,2,1,2] Output: 4
 - ◆ Given two positive integers a and b, the task is to find the GCD of the two numbers. Input: a=20, b=28 Output:4